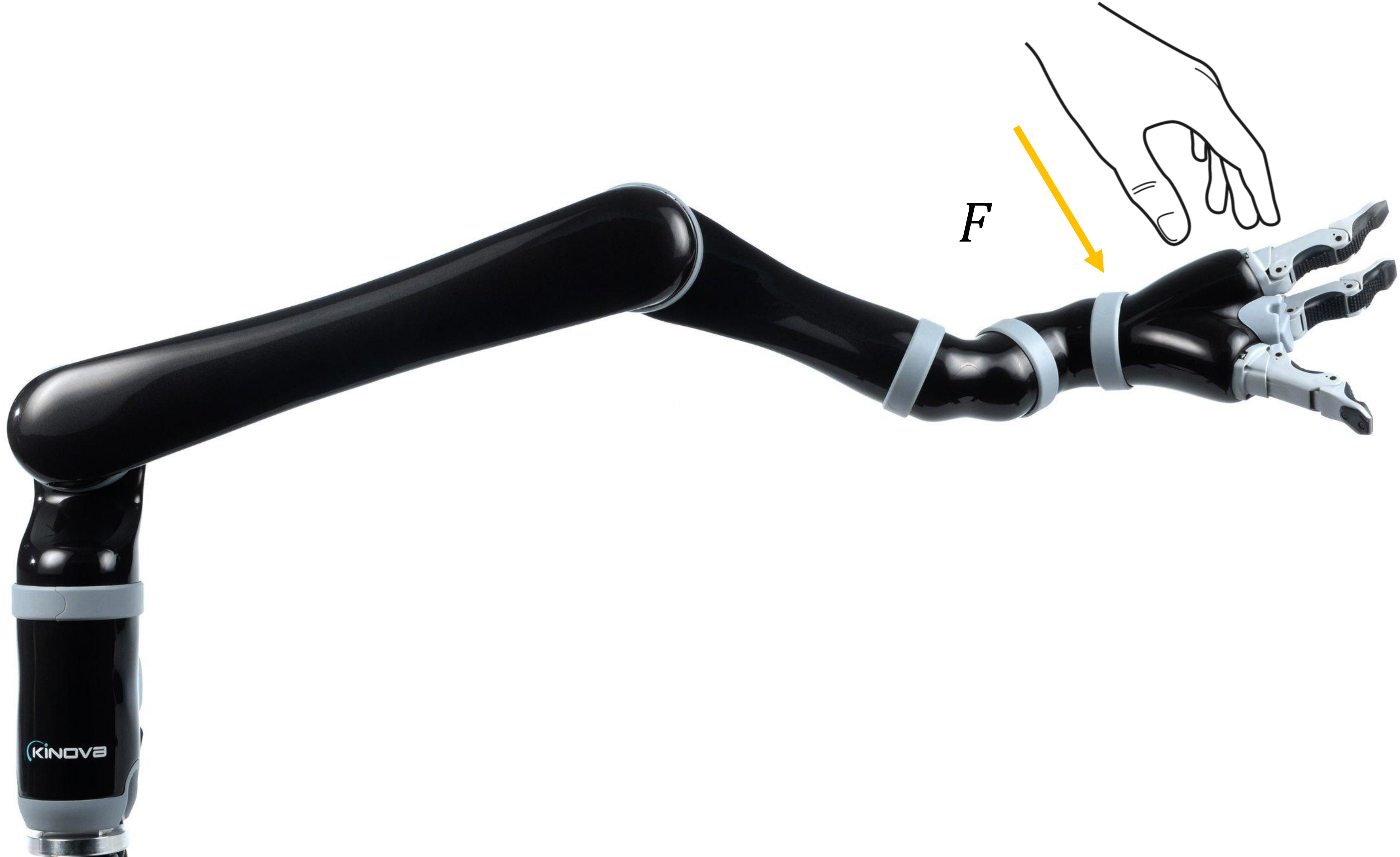
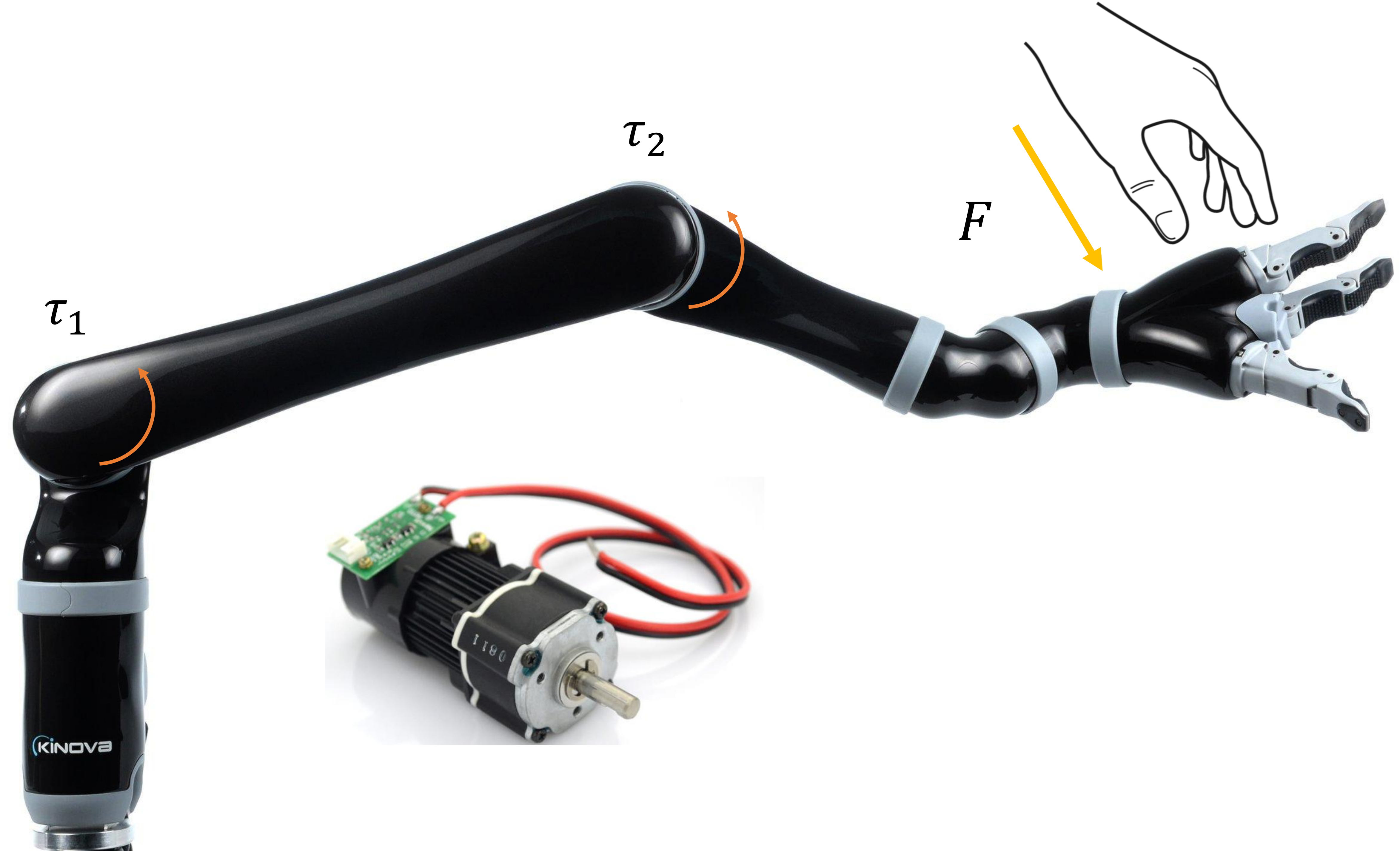
Physical Interaction

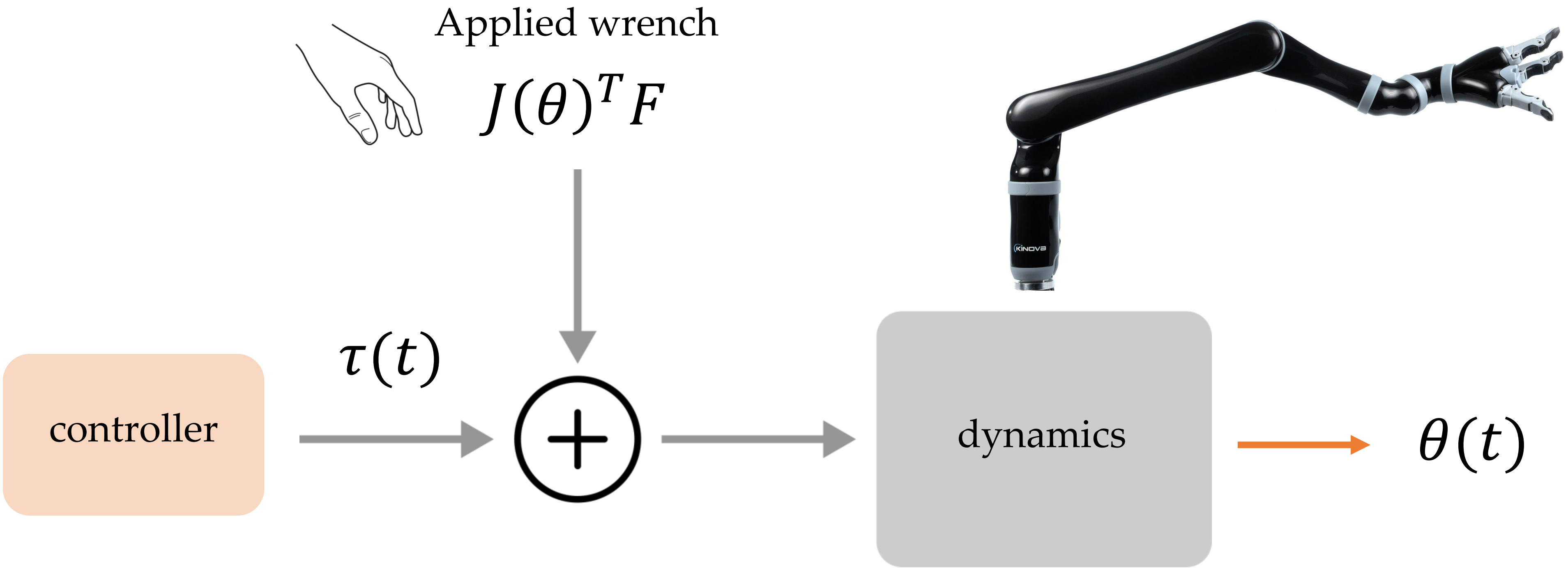
Dylan Losey

This Lecture

- Formulating physical interaction
- Outlining response strategies


F


τ_1
τ_2
F
KINOVA



$$\underbrace{\tau + J(\theta)^T F}_{\text{motor torques, applied forces}} = \underbrace{M(\theta)\ddot{\theta} + C(\theta, \dot{\theta})\dot{\theta} + g(\theta)}_{\text{mass, inertia, friction, gravity}}$$

Detecting Interaction

Three common methods to determine when physical interaction is occurring:

1) *Sensors. Use a force/torque sensor or a camera. The information is localized to the sensor's position.*
2) *Disturbance observer. Predict where the robot should move based on the control commands. If the robot's motion deviates from this prediction, then you have a disturbance. Relies on accurate model.*
3) *Motor current. The current in DC motors is proportional to the torque; you will see current spikes when physical interaction occurs. Not very accurate, relies on specific actuation types.*

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How would you decide if a physical interaction is *intentional* or *unintentional?*

Detecting Interaction

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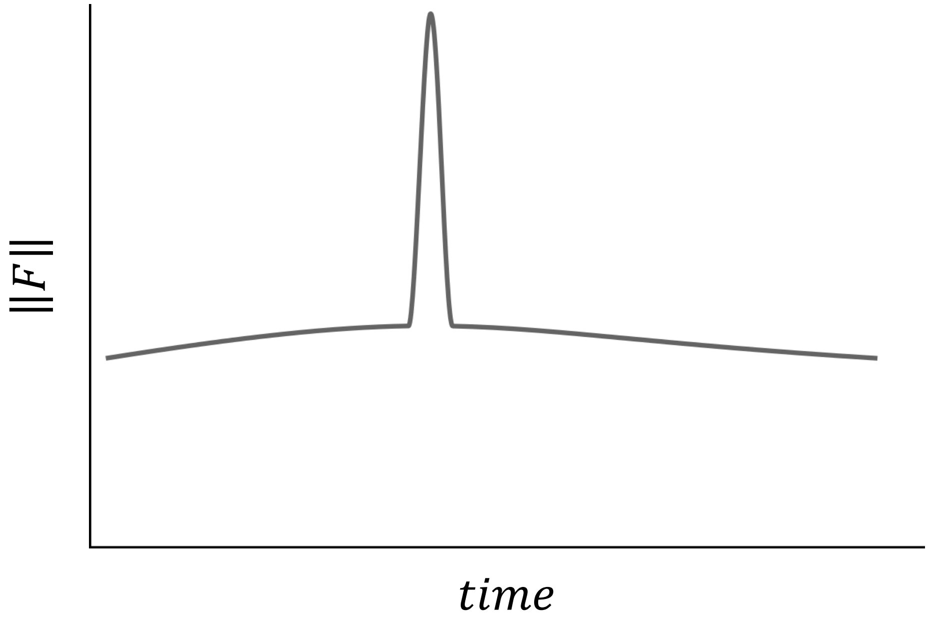
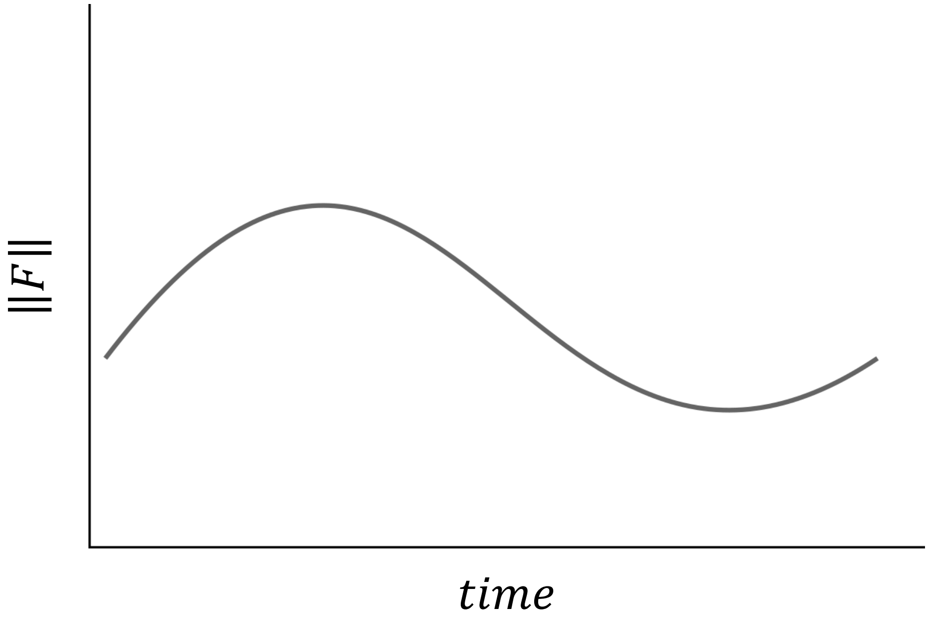
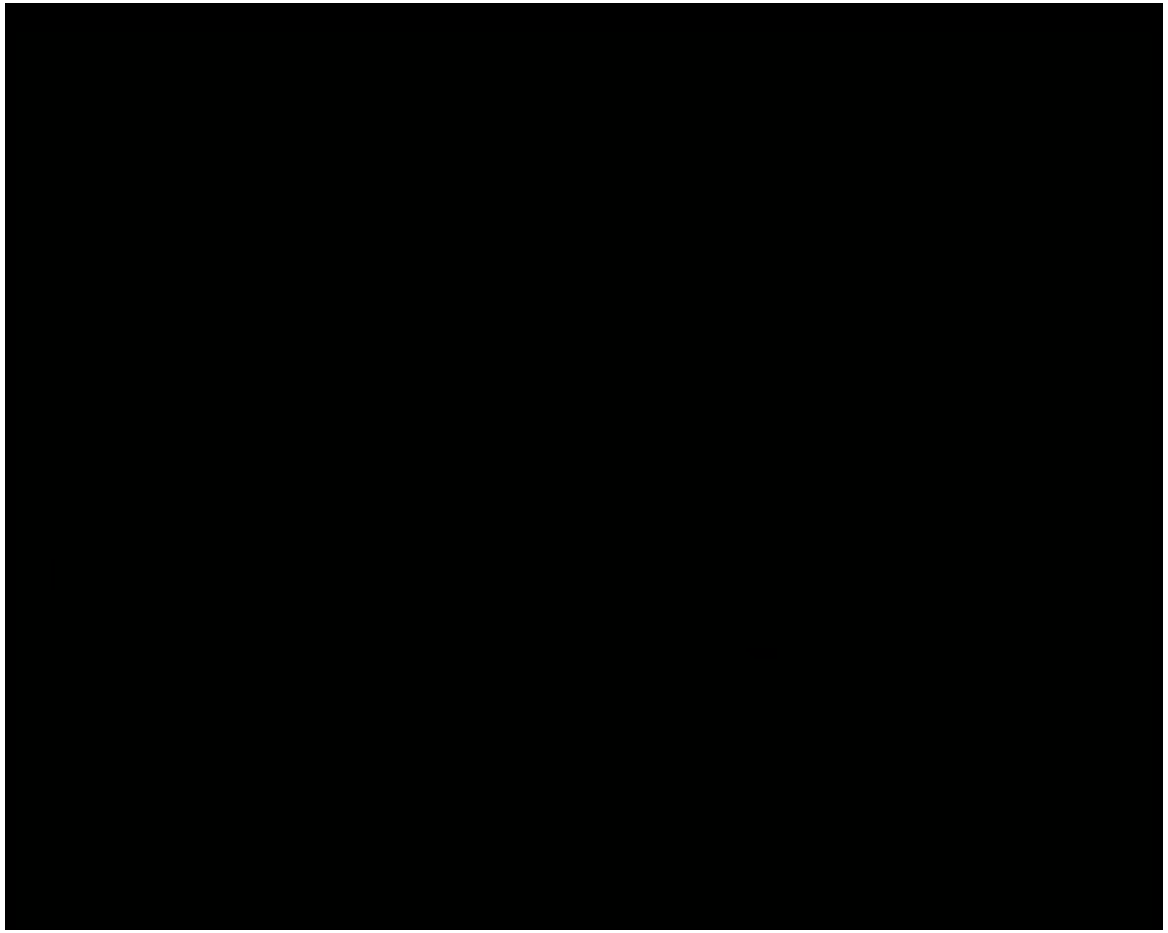






How should robots respond to physical interaction?

Soft Robots

We can make robots inherently safer during physical interaction by making them *softer*.

Soft Robots

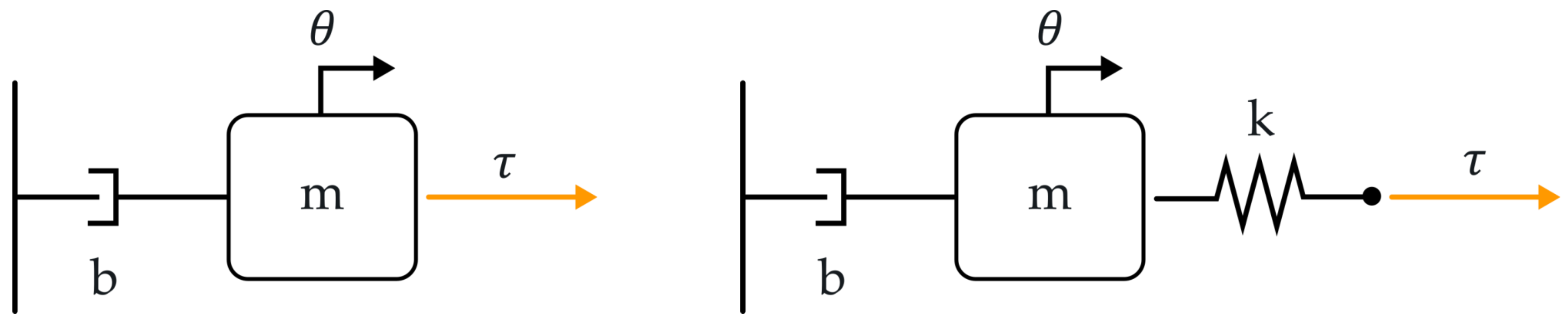
We can make robots inherently safer during physical interaction by making them *softer*.

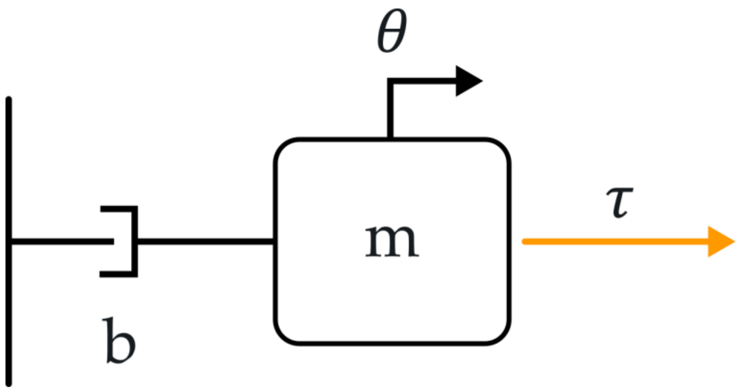
Can you think of some of the pros and cons with using soft robots? What tasks might they be good at, and where might they struggle?

Soft Robots

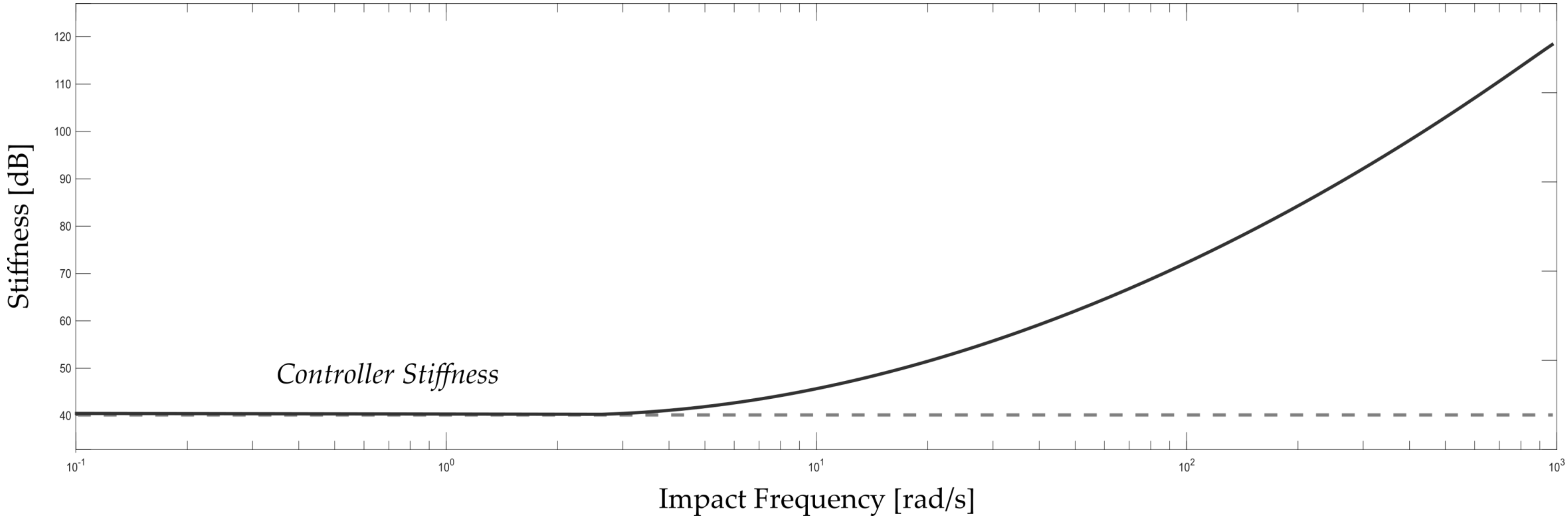
We can make robots inherently safer during physical interaction by making them *softer*.

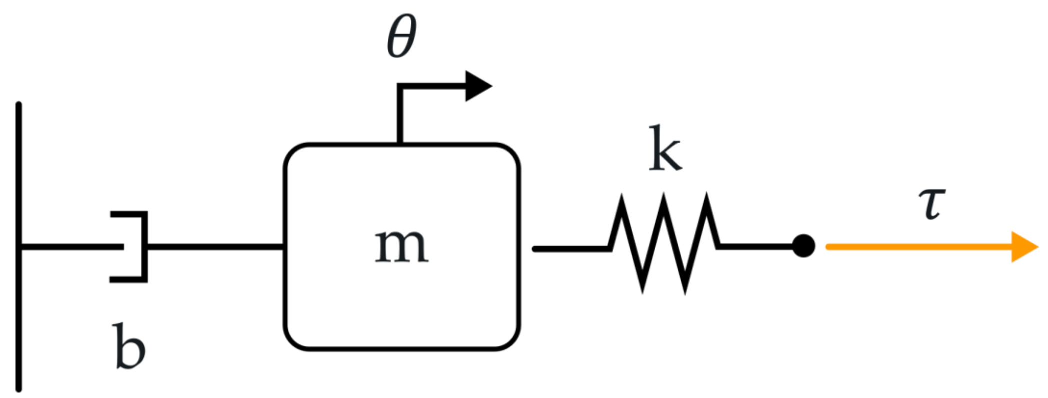
Soft robots take many forms, but we can illustrate how they work by comparing a mass-damper (1-DoF robot) with a **rigid or compliant transmission**.



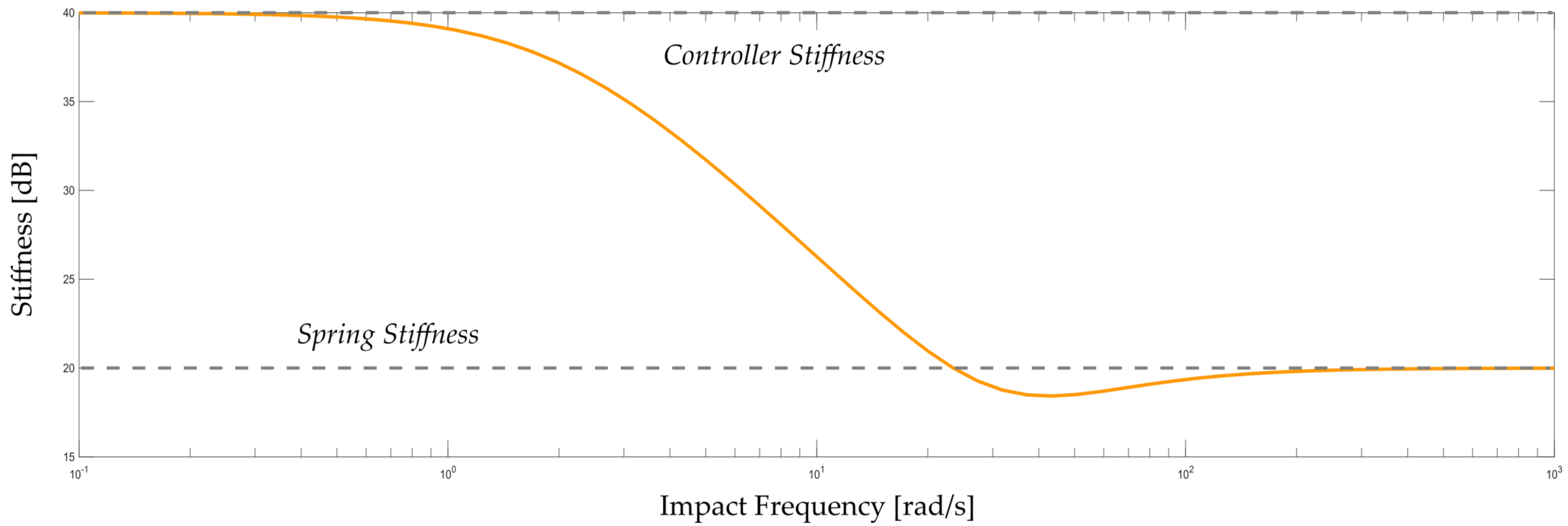


During sudden impacts, the robot behaves like a rigid link (high stiffness), regardless of the control strategy.



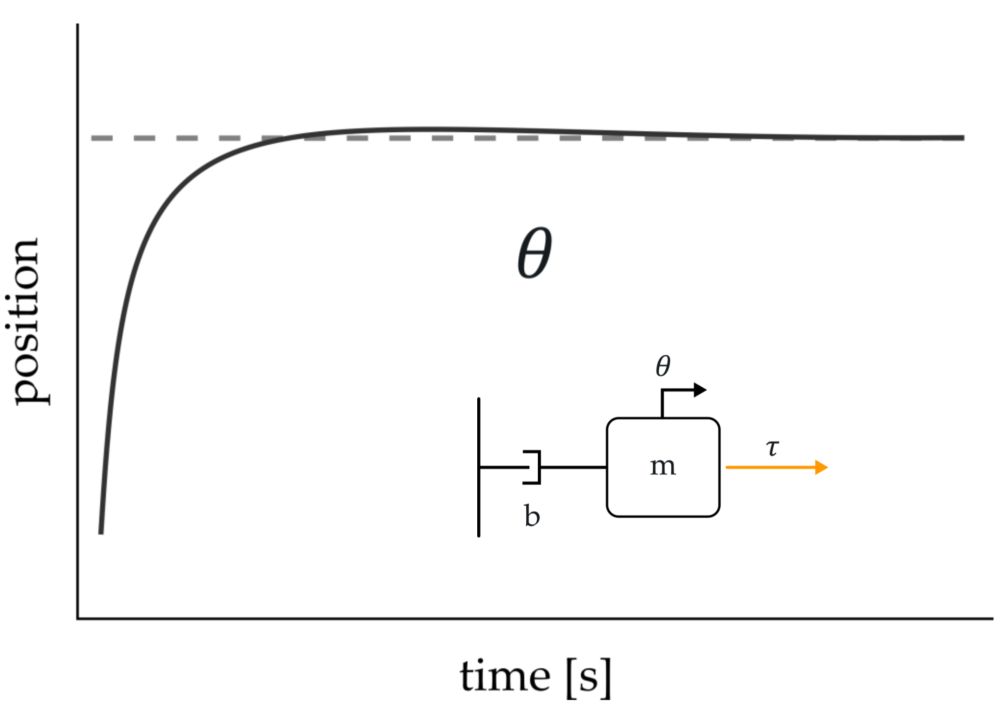


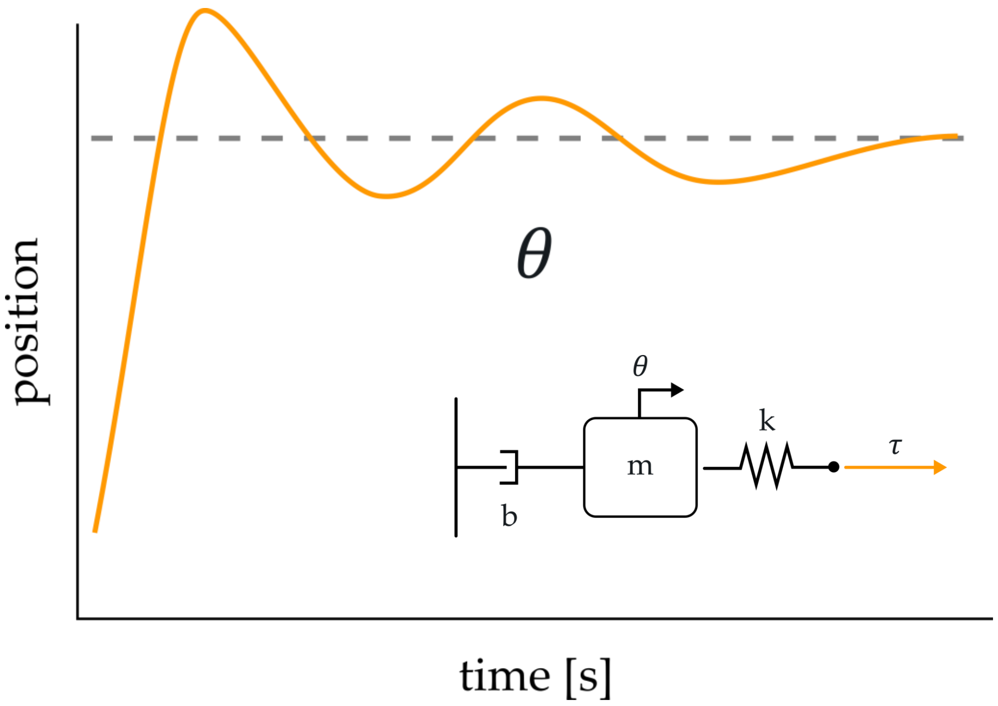
During sudden impacts, the robot behaves like a spring (rendering k), regardless of the control strategy.



Soft Robots

Adding mechanical compliance improves safety but *generally harms performance*. Here we see how adding compliance to the transmission leads to undesirable oscillations.





What control strategies help rigid robots with physical interaction?

Response Heuristics

(1) Stop moving when you sense a collision.

If torque $\tau > \delta$:

collision sensed

stop robot

Response Heuristics

(1) Stop moving when you sense a collision.

(2) Switch modes when you sense a collision.

If torque $\tau > \delta$:

collision sensed

set joint torque $\tau \leftarrow g(\theta)$

$$\cancel{\tau} = M(\theta)\ddot{\theta} + C(\theta, \dot{\theta})\dot{\theta} + \cancel{g(\theta)}$$

Response Heuristics

(1) Stop moving when you sense a collision.

(2) Switch modes when you sense a collision.

(3) Rescale trajectory when you sense a collision

While trajectory ξ not complete:

Move towards state $s^t \leftarrow \xi^t$

If torque $\tau < \delta$:

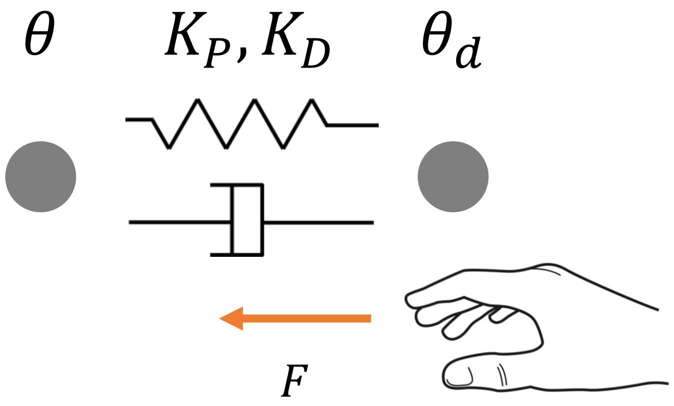
$t \leftarrow t + 1$

Wait to advance to next state until no collision


HIGH
VIRTUAL
SPRING RATE

Impedance Control

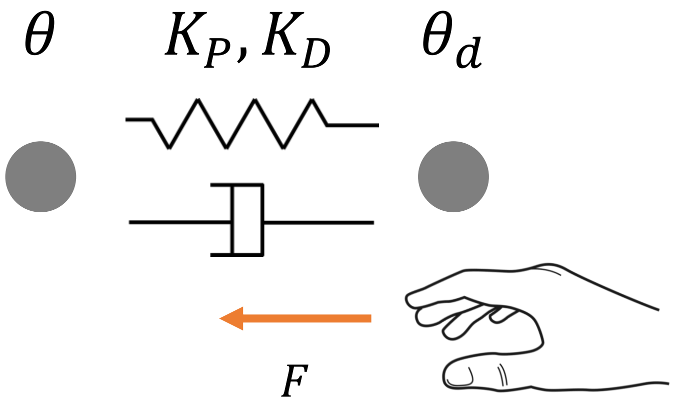
Impedance control makes the robot behave like a **mass-spring-damper** with stiffness, damping, and equilibrium of our choice.



Impedance Control

Impedance control makes the robot behave like a **mass-spring-damper** with stiffness, damping, and equilibrium of our choice.

We choose stiffness matrix K_P, damping matrix K_D, and desired trajectory $\theta_d(t)$, $\dot{\theta}_d(t)$



Impedance Control

$$\tau = K_D\left(\dot{\theta}_d - \dot{\theta}\right) + K_P(\theta_d - \theta)$$

Increasing K_D and K_P **increases the impedance**:
the human must apply large external forces to move the robot

Decreasing K_D and K_P **decreases the impedance**:
the human can backdrive or guide the robot with small forces

Can prove that this control strategy is stable for positive gains.

How do we apply impedance control to common physical interaction settings?


3/320
OT
ASUS

Assist-as-Needed Control

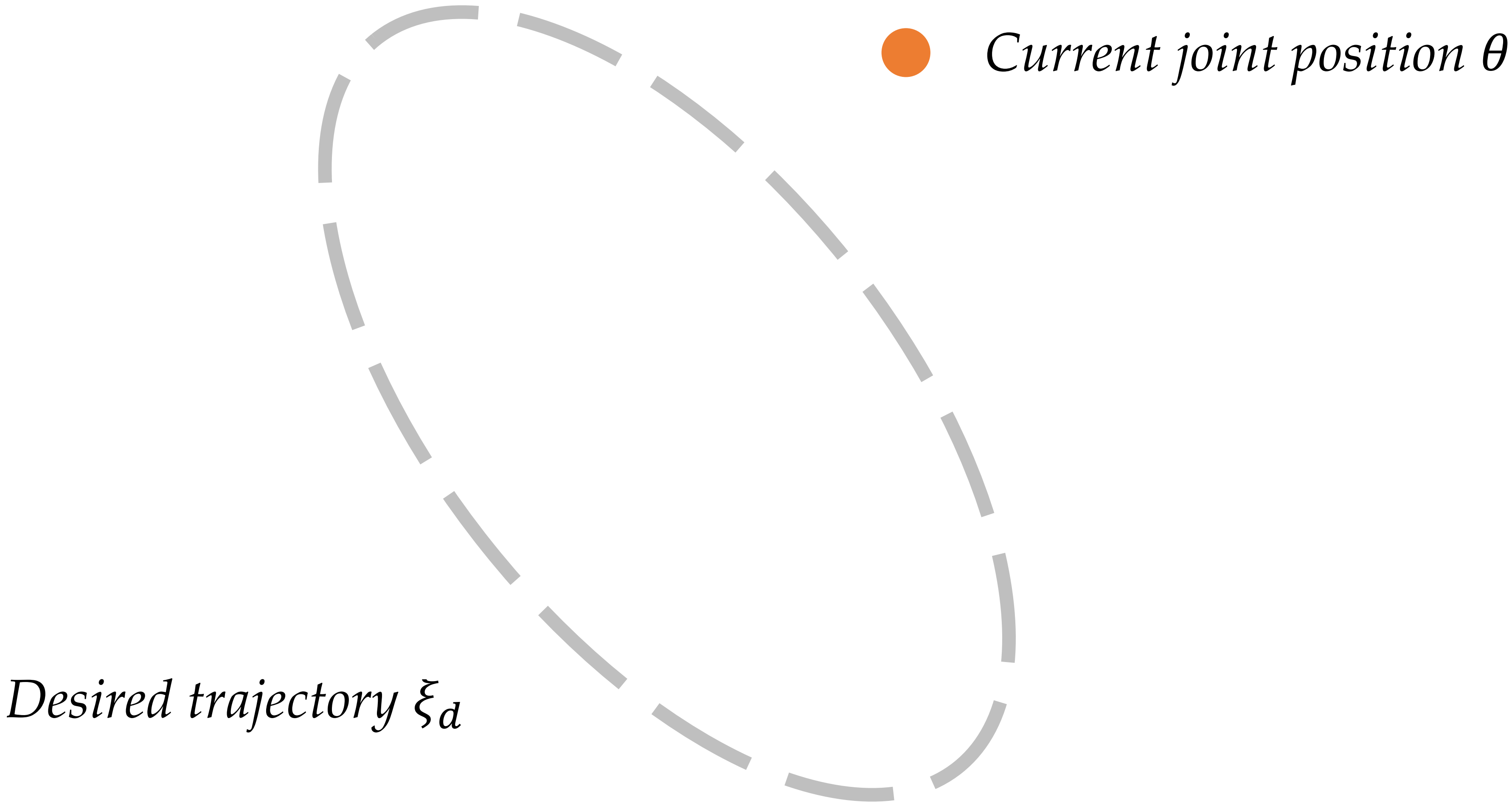
Consider a system with desired trajectory ξ_d

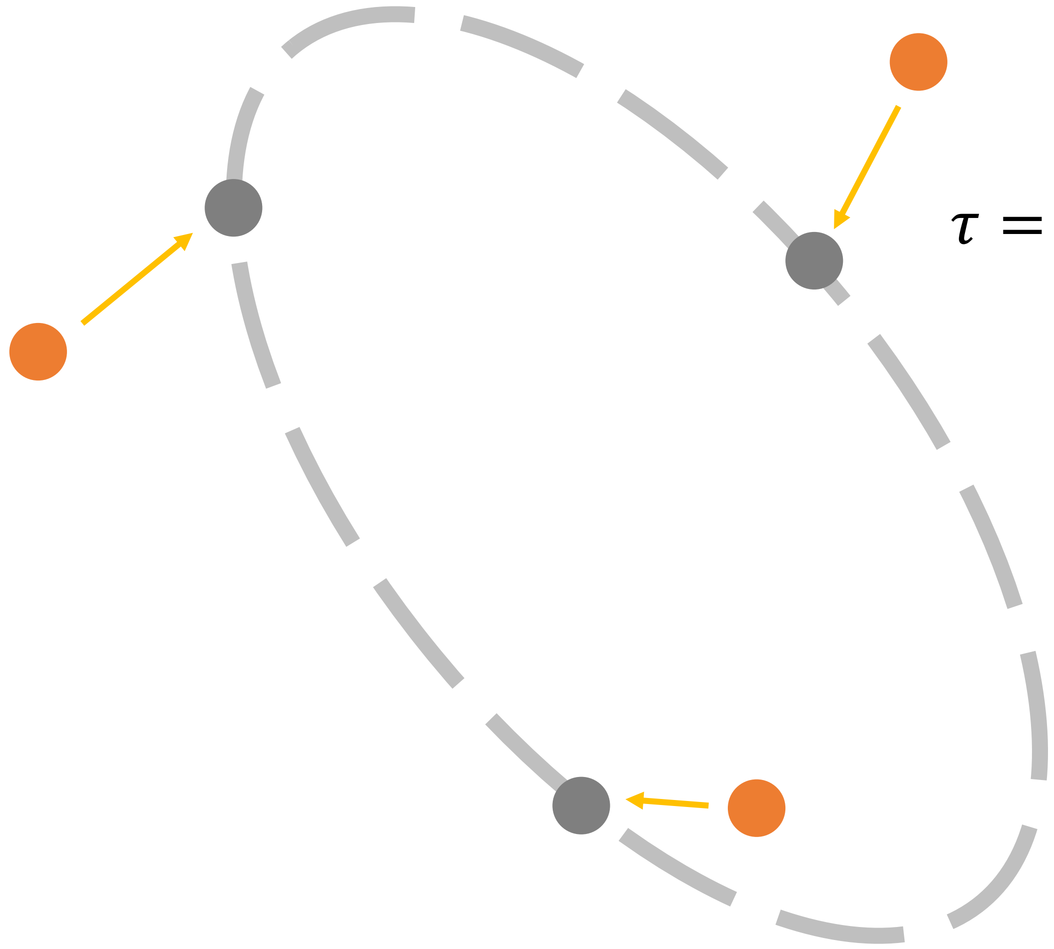
The robot expert thinks this is the best possible trajectory.

$$\tau = K_p\left(\xi_d^t - \theta\right)$$

stiffness gain *joint position*

desired trajectory


Current joint position θ
Desired trajectory ξ_d



$$\tau = K_p\left(\xi_d^t - \theta\right)$$

Assist-as-Needed Control

$$\tau = K_p\left(\xi_d^t - \theta\right)$$

Increasing K_p
Robot is in charge. Human can barely deviate from ξ_d

Decreasing K_p
Human can make larger mistakes. When human intervenes, they lead

Assist-as-Needed

Initialize gain K_p^0 and desired trajectory ξ_d

For each iteration i

For each timestep t

$\theta^t \leftarrow$ current joint position

$\tau \leftarrow K_p^i\left(\xi_d^t - \theta^t\right)$ calculate corrective torque

$\theta^{t+1} \leftarrow$ dynamics with τ and human wrench

Assist-as-Needed

Initialize gain K_p^0 and desired trajectory ξ_d

For each iteration i

For each timestep t

$\theta^t \leftarrow$ current joint position

$\tau \leftarrow K_p^i(\xi_d^t - \theta^t)$ calculate corrective torque

$\theta^{t+1} \leftarrow$ dynamics with τ and human wrench

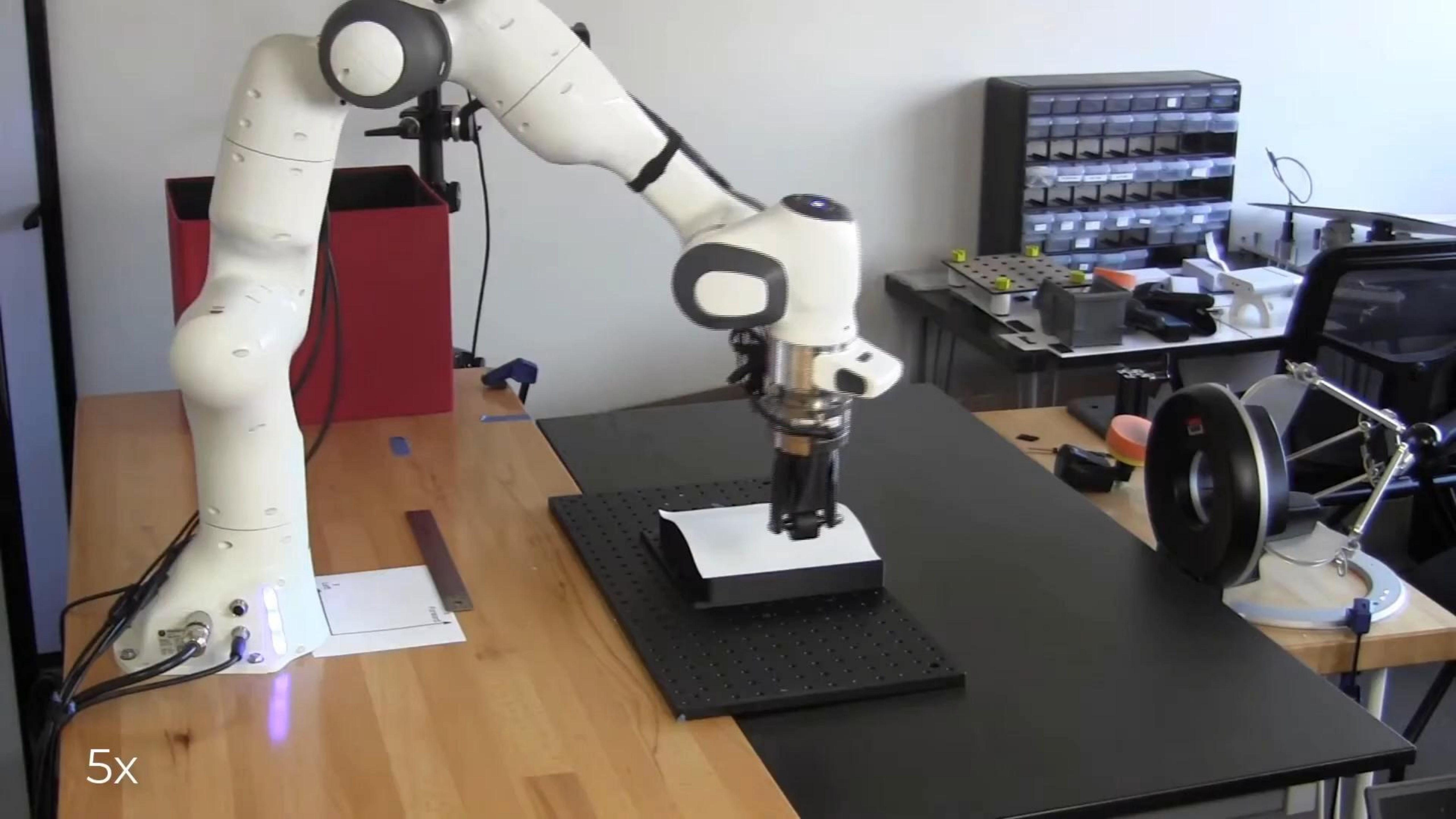
$\xi \leftarrow$ actual trajectory followed

$e^i \leftarrow \|\xi - \xi_d\|$ find task error

if $e^i < \delta$ then $K_p^{i+1} \leftarrow K_p^i - \gamma$

if $e^i > \delta$ then $K_p^{i+1} \leftarrow K_p^i + \gamma$

Decrease assistance if performance is below threshold;
Otherwise provide more help

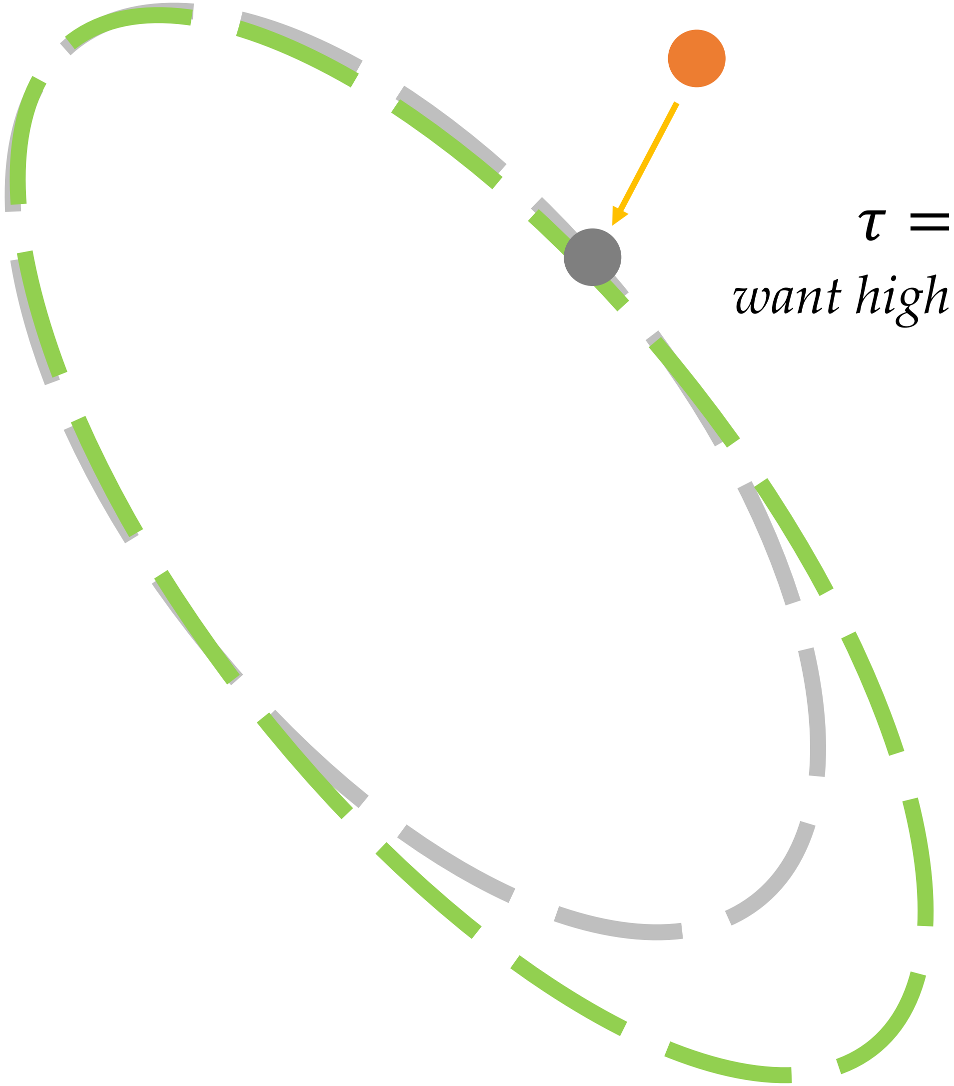

5x

Current joint position θ

In **shared control**, we want to enable humans to modify the robot's motion

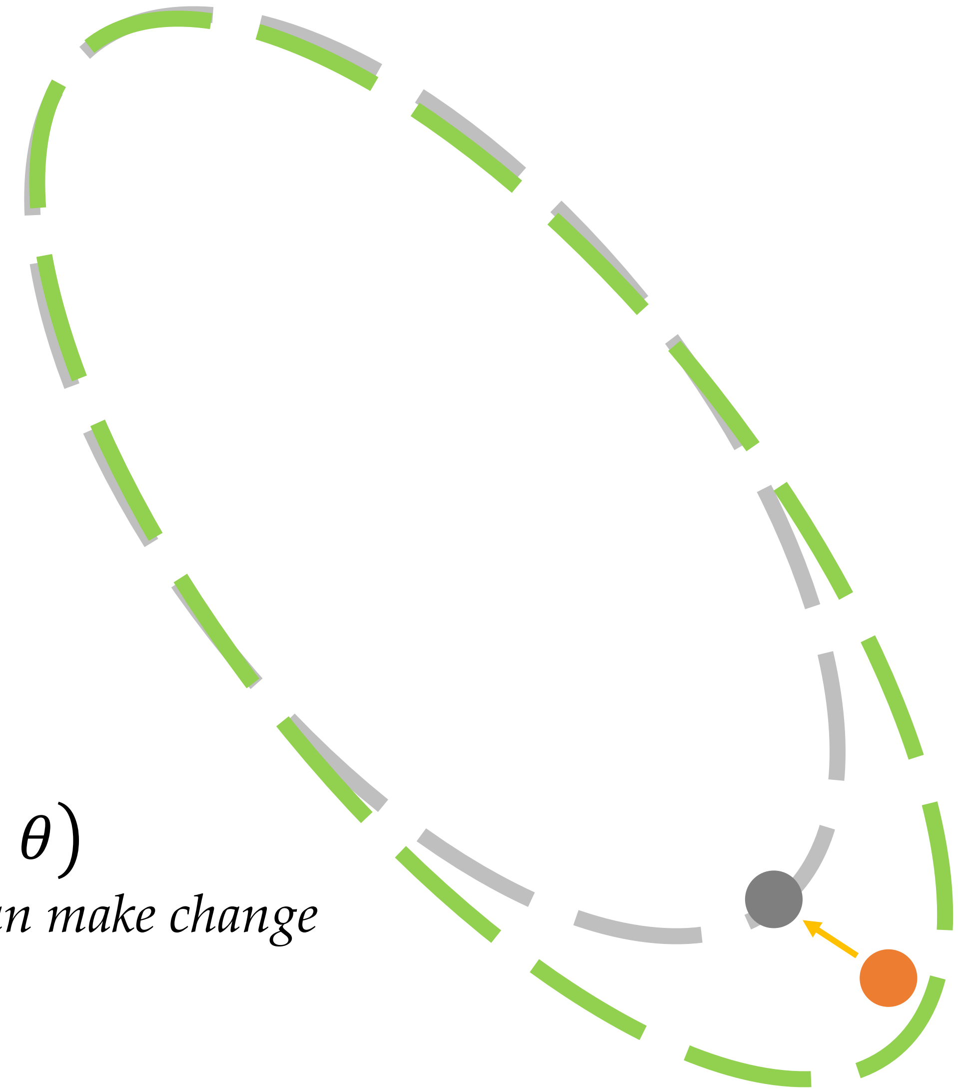
Desired trajectory ξ_d

User's preferred trajectory



$$\tau = K_p\left(\xi_d^t - \theta\right)$$

want high K_p to automate trajectory



$$\tau = K_p\left(\xi_d^t - \theta\right)$$

want low K_p so human can make change

Shared Control

Initialize gain K_p^0 and desired trajectory ξ_d

For each timestep t

$\theta^t \leftarrow$ current joint position

$F^t \leftarrow$ wrench (force and torque) applied by human

$\tau \leftarrow K_p^t(\xi_d^t - \theta^t)$ calculate corrective torque

$\theta^{t+1} \leftarrow$ dynamics with τ and human wrench

if $\|F^t\| > \delta$ then $K_p^{t+1} \leftarrow K_p^t - \gamma$

if $\|F^t\| < \delta$ then $K_p^{t+1} \leftarrow K_p^t + \gamma$

If human applies large wrench arbitrate control to human;
Otherwise increase gain to make robot more accurate

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- Formulating physical interaction
- Outlining response strategies

Next Lecture

- Modeling humans so we can learn from their actions